

Some results on Interval probe graphs

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Abstract

Interval Probe Graphs a generalization of Interval Graphs, were first introduced in 1994. In this paper, we found some relations among Interval Probe Graphs, Proper Probe Graphs and Interval Graphs. By studying Trees and Chordal Graphs. Moreover, We also characterize the differences between Interval Graphs and Proper Probe Graphs.

intersection graph of that same type, where $V(G_X) = X \subseteq V(G)$ and $E(G_X) = \{uv \in E(G) \mid u, v \in X\}$.

An interval graph G that has a representation in which each interval has the same length is called a unit interval graph. Similarly, if G has a representation in which no interval properly contains another interval, G is called a proper interval graph

1 Introduction

In this chapter, we will provide the formal definition of a Interval probe graph, and also give a brief review of many of the important families of graphs which are related in some way to interval probe graphs.

1.1 Interval graphs

A graphs are important for their applications to scheduling problems, microbiology, and VLSI circuit design. The interval graph for this example is shown in Figure 1.1. Finding a consistent assignment of rooms can be viewed as a coloring problem on the interval graph, where the meeting rooms are the colors and adjacent vertices must be assigned different colors. There are efficient algorithms for coloring the vertices of an interval graph using a minimum number of colors [1]. In our example, there cannot be a solution with four rooms since the interval graph has a clique (or complete subgraph) of size five. Indeed, the only subsets that could be colored by the same color in this example are $\{1, 4\}$ or $\{1, 5\}$ or $\{1, 6\}$. A stable set (or independent set) is a subset of vertices no two of which are connected by an edge. Here there is no stable set larger than size 2. All of these families of intersection graphs satisfy the hereditary property, namely, if a graph $G = (V, E)$ is the intersection graph of a certain type (e.g., intervals, trapezoids, etc.), then every induced subgraph G_X of G is also an

1.2 Tolerance graphs

A graph $G = (V, E)$ is a tolerance graph if each vertex $v \in V$ can be assigned a closed interval I_v and a tolerance $t_v \in \mathbb{R}^+$ so that $xy \in E$ if and only if $|I_x \cap I_y| \geq \min\{t_x, t_y\}$. Such a collection (I, t) of intervals and tolerances is called a tolerance representation where $I = \{I_x \mid x \in V\}$ and $t = \{t_x \mid x \in V\}$. If graph G has a tolerance representation with $t_v \leq |I_v|$ for all $v \in V$, then G is called a bounded tolerance graph and the representation is called a bounded tolerance representation. See Figure 1.2.1 and Figure 1.2.2.

1.3 Chordal graphs and AT-free

Graphs

A graph G is a chordal graph if every cycle of length greater than or equal to 4 has a chord, that is, an edge connecting two vertices that are not consecutive on the cycle. For example, the graph in Figure 1.1 is chordal, and the edge $(3, 5)$ is a chord of the cycle $[3, 4, 5, 6, 3]$. The chordal graphs are a well known classical family of graphs, and they appear in many interesting applications including relational databases, matrix theory, statistics and biology. In the literature, chordal graphs are also called triangulated graphs or rigid circuit graphs. The family of chordal graphs includes all interval graphs but does not include all tolerance graphs.

A graph G is asteroidal triple if three vertices of a graph form an asteroidal triple if every two of them are connected by a path

avoiding the neighbourhood of the third. A graph is AT-free graph if it does not contain any asteroidal triple.

Theorem 1.3 A chordal graph G is AT-free if and only if it does not contain the graphs $A_1, A_2, B_n (n \geq 1), E_n (n \geq 1)$ as an induced subgraph. as shown in Figure 1.3.1

1.4 Interval probe graphs

An undirected graph $G = (V, E)$ is an interval probe graph if the vertex set can be partitioned into two subsets, P (probes) and N (non-probes), where N is a stable set and there is a completion $E' \subseteq \{uv \mid u, v \in N, u \neq v\}$ such that $G' = (V, E \cup E')$ is an interval graph. Equivalently, G is an interval probe graph if we can assign an interval to each vertex such that two vertices are adjacent if and only if at least one of them is in P and their corresponding intervals intersect. We say that E' is an interval completion of G on N . We remark that it is possible that a graph can be a probe graph with respect to different partitions of probes and non-probes, giving very different interval completions. Clearly, the interval probe graphs constitute a generalization of interval graphs (Where $N = \emptyset$).

An interval probe graph which is not an interval graph is shown in Figure 1.4.1 using the probe/non-probe partition $P = \{a, b, c, d\}$ and $N = \{x, y\}$. If the edge xy is added to the graph, we can construct an interval representation also shown in Figure 1.4.2 where the intervals representing probe vertices are drawn with thick line.

2 Relations of probe graphs and interval graphs

2.1 Relations of interval probe and trees

Theorem 2.1.1 In trees, a graph T is an interval probe graph if and only if T does not contain T_3 .

Proof:

$(\Rightarrow)$

T is an interval probe graph.

$\Rightarrow T$ is a tolerance graph.[1]

$\Rightarrow T$ does not contain T_3 . [3]

$(\Leftarrow)$

T does not contain T_3 . The model of subgraph is shown in Figure 2.11.

$\Rightarrow T$ is an interval probe graph

Theorem 2.1.2 In trees, a graph T is a proper probe graph if and only if T does not contain T_2 .

Proof:

$(\Rightarrow)$

T is an proper probe graph.

$\Rightarrow T$ is a bounded tolerance graph.[1]

$\Rightarrow T$ does not contain T_2 . [3]

$(\Leftarrow)$

T does not contain T_2 . The model of subgraph is shown in Figure 2.12.

$\Rightarrow T$ is a proper probe graph

2.2 Relations of interval probe graphs and interval graphs

Lemma 2.2.1 The following conditions are equivalent:[4]

(i) G is an interval graph.

(ii) G is chordal and AT-free.

Theorem 2.2 For chordal graphs, if a graph G is an Interval Probe Graph which does not contain $\{A_1, A_2, B_n (n \geq 1), E_n (n \geq 1)\}$, then G is an Interval Graph.

Proof:

In Figure 2.2.1~ Figure 2.2.4, we know $\{A_1, A_2, B_n, E_n\} \in \text{Interval Probe}$. G is a chordal graph and AT-free. By Lemma 2.2.1, G is an Interval Graph.

We can discuss the Hierarchy of Interval and Interval Probe in Figure 2.2.5

2.3 Difference between Interval graphs and proper probe graphs

Lemma 2.3.1 If G is a proper probe graph.

$\Rightarrow G$ is a bounded tolerance graph.

$\Rightarrow G$ is an AT-free graph.[5][6]

$\Rightarrow G$ does not contain $C_n (n \geq 5)$. [1]

Theorem 2.3.1 If a proper probe graph G does not contain $C_n (n \geq 4)$. Then G is an interval graph.

Proof:

G does not contain $C_n, n \geq 4$. G is a

chordal graph. By Lemma 2.3.1, G is also an AT-free graph. So G is an interval graph.

Theorem 2.3.2 If an interval graph G does not contain Y , then G is a proper probe graph.

Proof:

An interval graph G is a proper interval graph if and only if it contains $K_{1,3}$. Now, since G is an interval graph, if G contains no $K_{1,3}$ then G is a proper interval graph and hence a proper probe graph. So we may assume there is a $K_{1,3}$ in G .

1)

In the subgraph of an interval graph, we can always pick the root vertex x as P . Because if x is N , then the interval representation is not a proper probe graph as in Figure 2.3.1

2)

There must be at least three non-adjacent groups of vertices connected to x . For each group, if there is an edge, then there must be a P in either vertex of that edge as in Figure 2.3.2.

3)

Now, if there is only one P , then the interval representation of this graph can be expressed as in Figure 2.3.3 and hence become a proper probe graph

4)

So assume at least two groups have a P . Then it is not a proper probe graph as shown in Figure 2.3.4.

4 Conclusion

In this paper, we figured out some results in Probe Graphs. For a tree T , T is an interval probe graph if and only if T does not induce T_3 . Also a graph T is a proper probe graph if and only if T does not induce T_2 . We proved that for a chordal graph G , if G is an Interval Probe Graph and contains no $\{A_1, A_2, B_n, E_n\}$ then G is an Interval Graph. We also characterized the difference between Interval Graphs and Proper Probe Graphs. If a Proper Probe Graph does not contain C_n ($n \geq 4$), then it will be an Interval Graph. And

last, for an Interval Graph G does not contain Y , then G is a Proper Probe Graph Fig. 2.4

5 References

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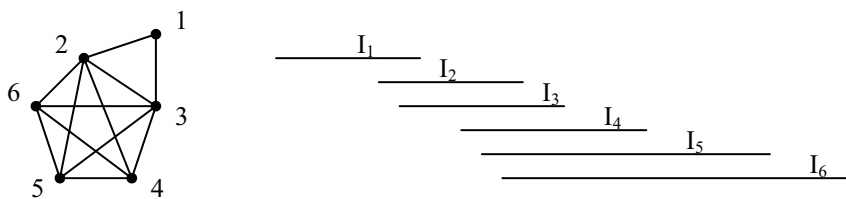


Figure 1.1

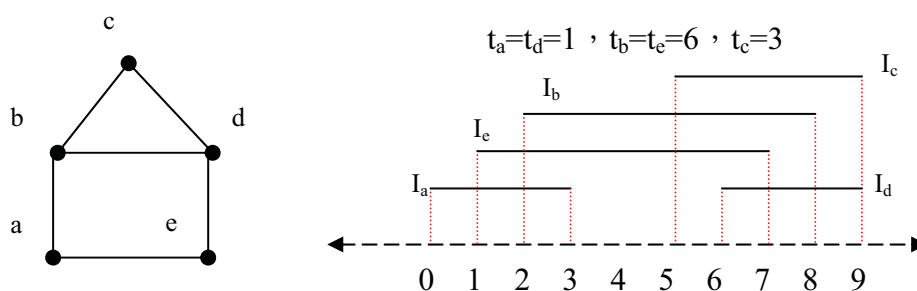


Figure 1.2.1

1.

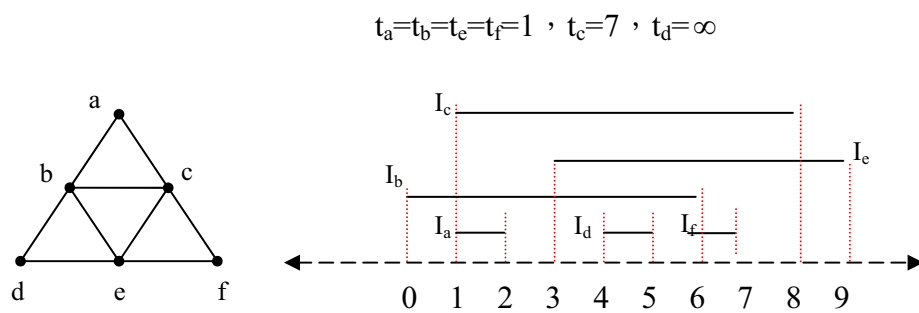


Figure 1.2.2

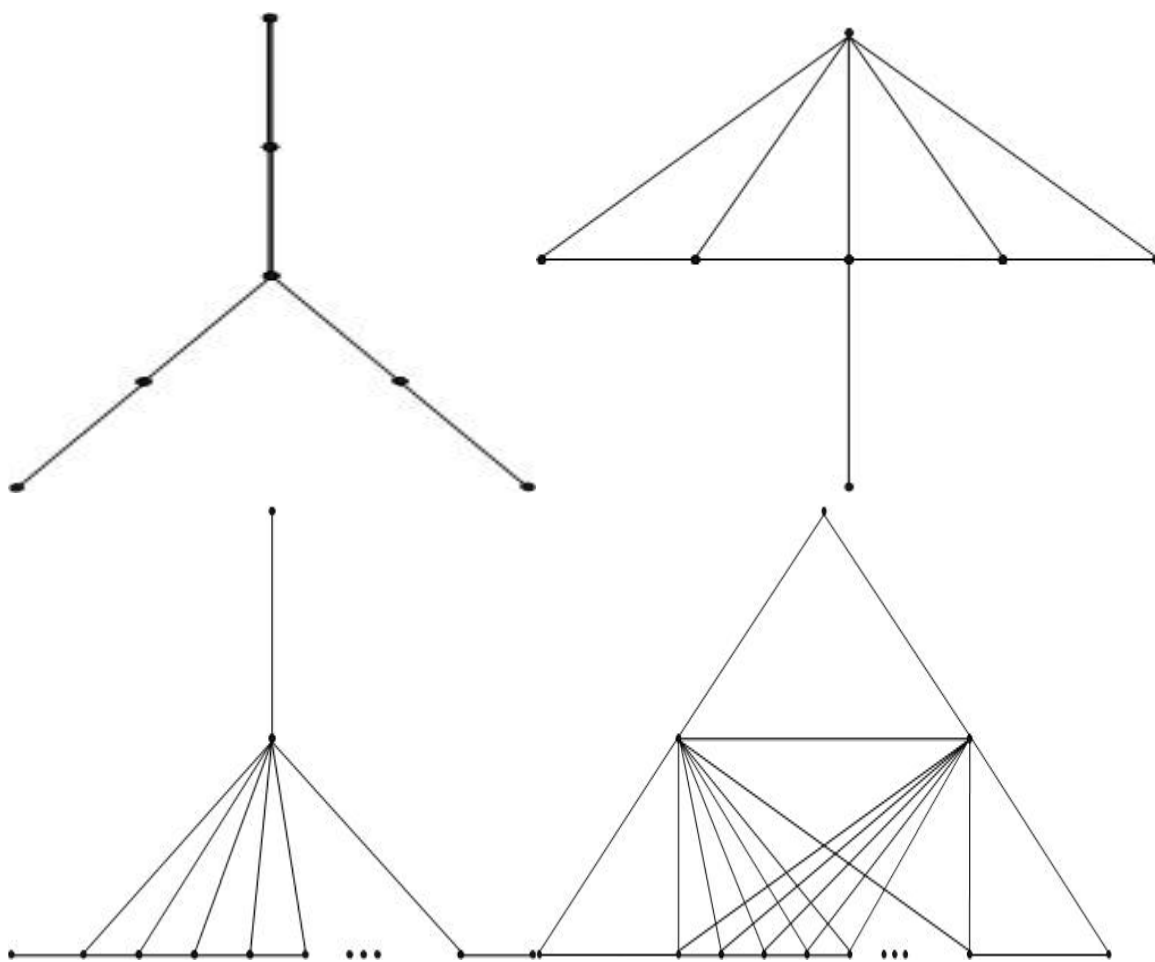


Figure 1.3.1

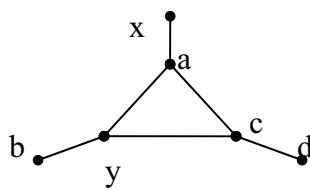


Figure 1.4.1

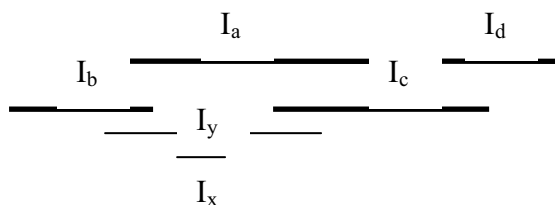


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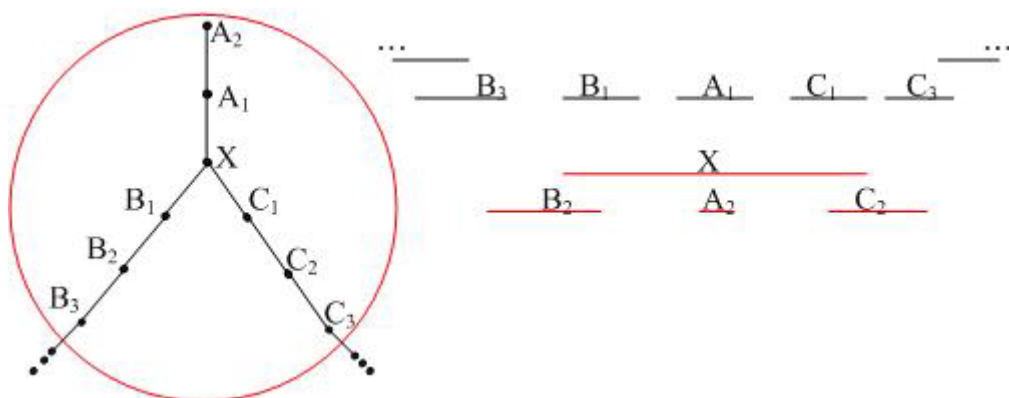


Figure 2.1.1

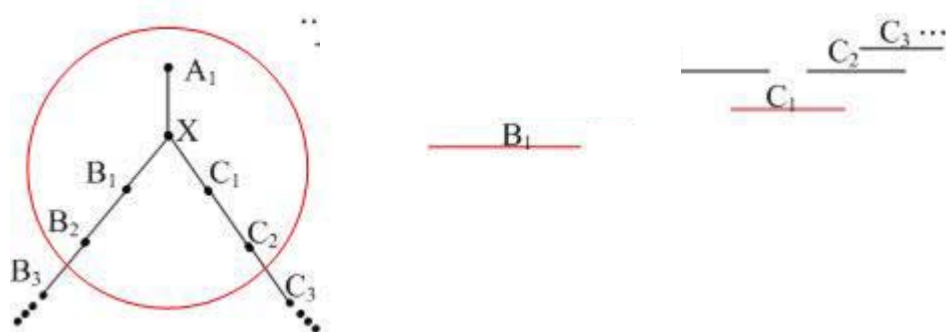


Figure 2.1.2

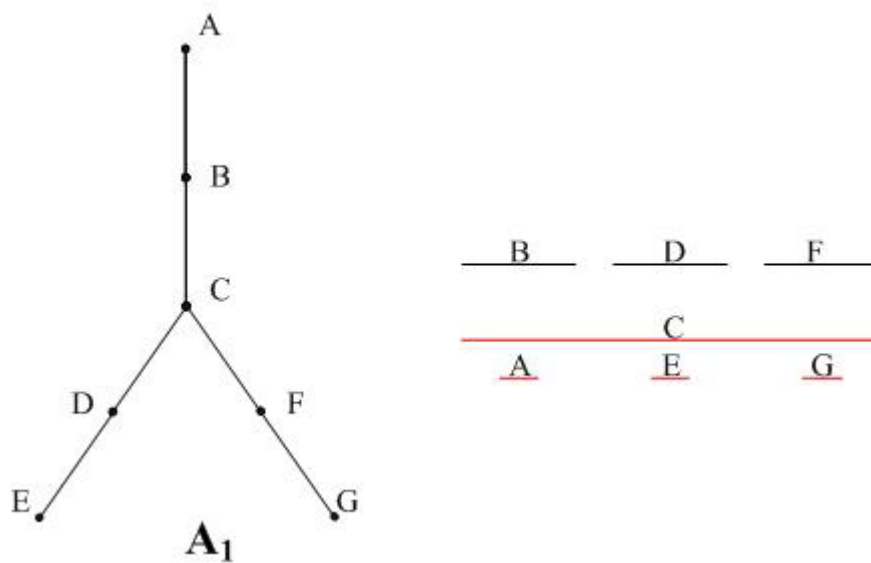


Figure 2.2.1

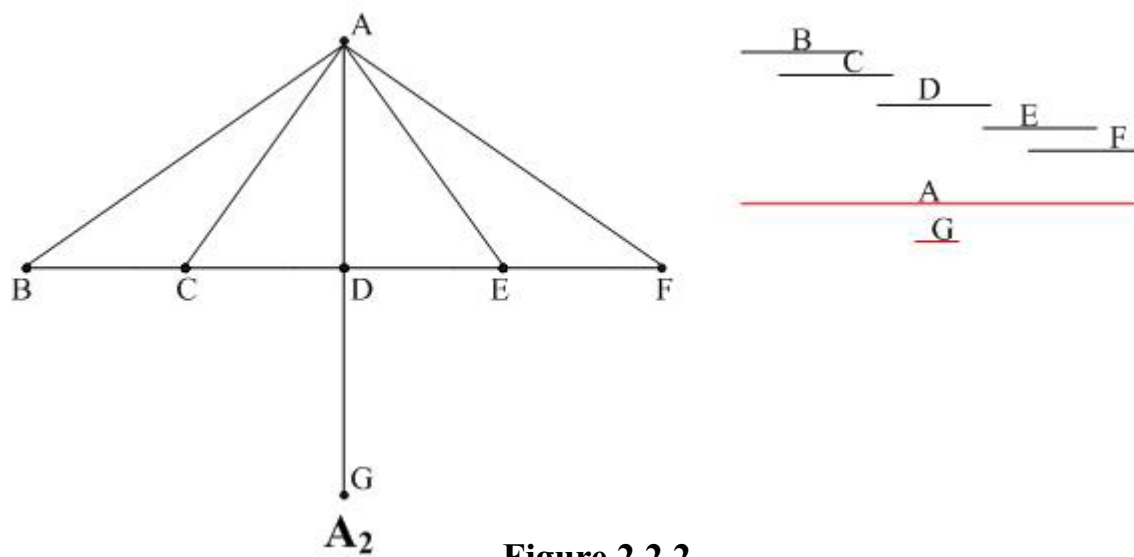


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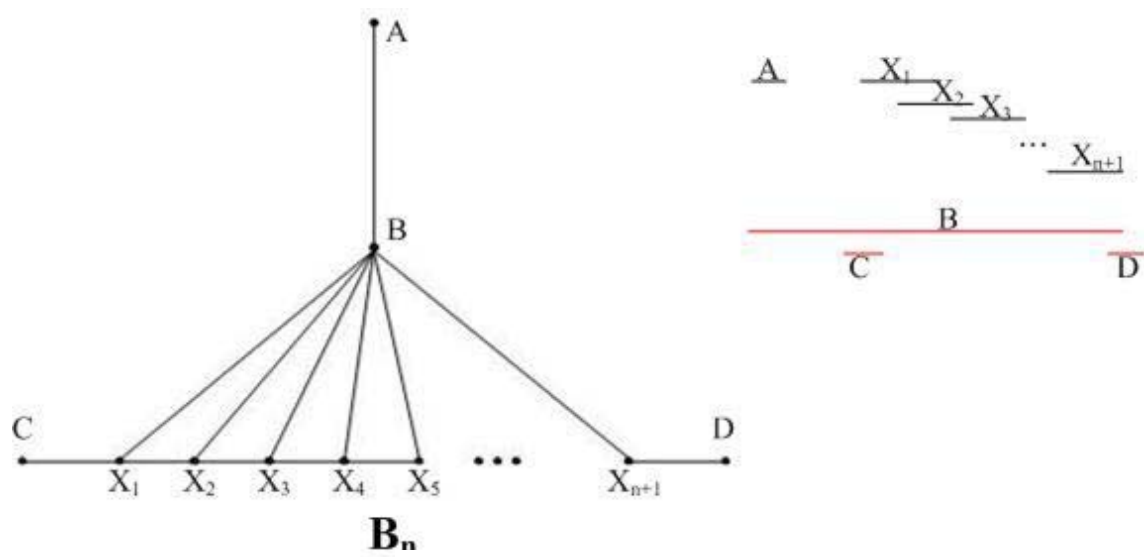


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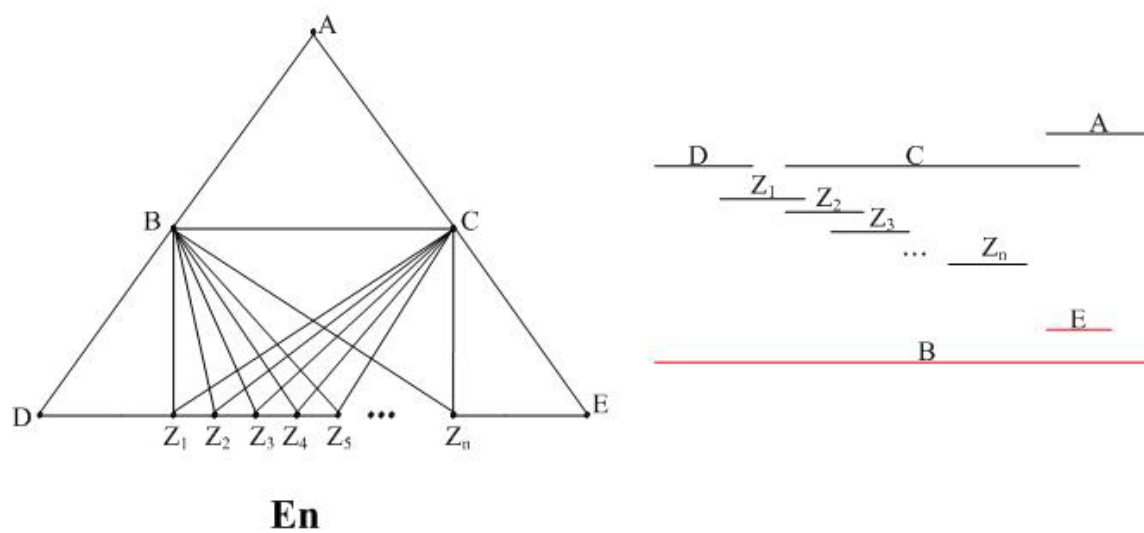


Figure 2.2.4

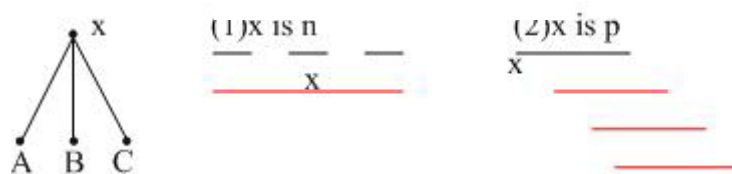


Figure 2.3.1

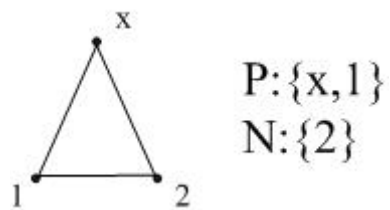


Figure 2.3.2



Figure 2.3.3

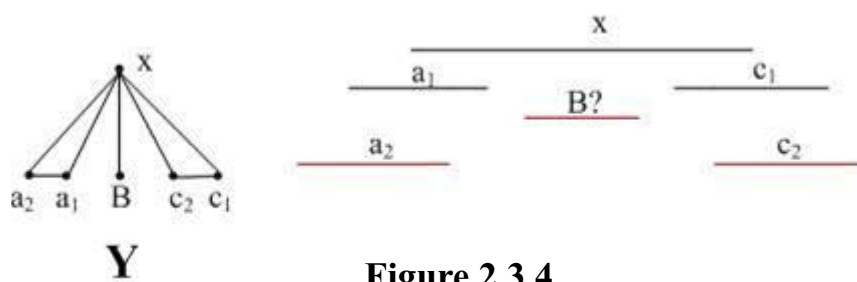


Figure 2.3.4

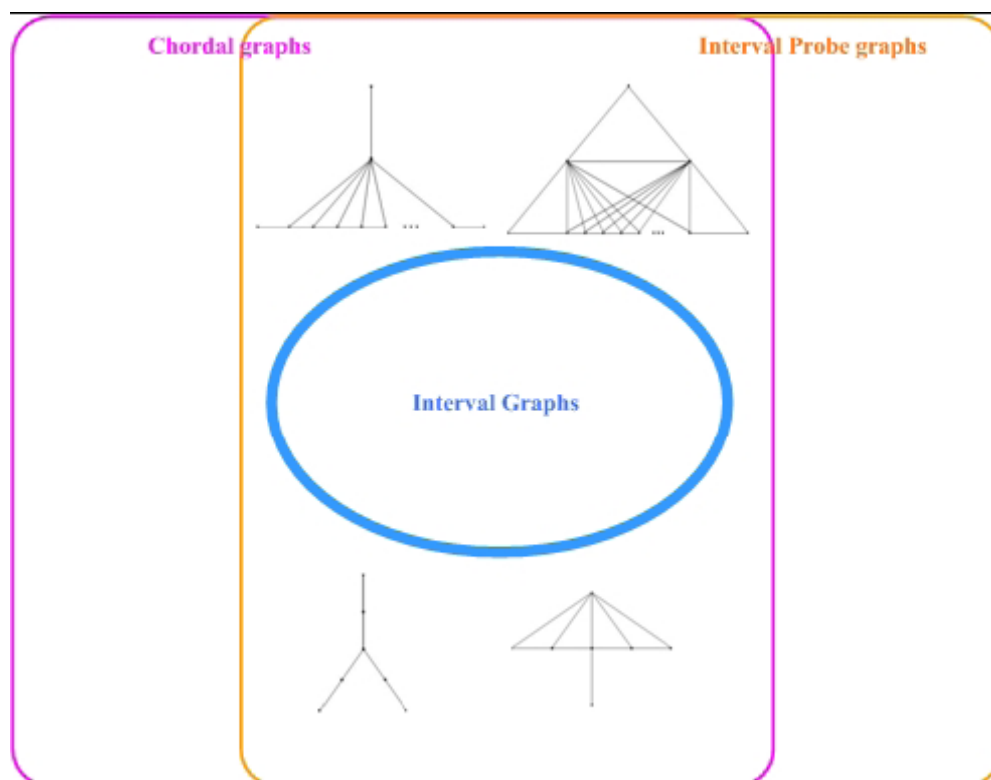


Figure 2.4